INCH-POUND

MIL-DTL-82960(OS)
23 October 1996

DETAIL SPECIFICATION

# INITIATOR, PROPELLANT ACTUATED, JAU-27/A (MF72)

*This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the requirements for the manufacture and acceptance of the JAU-27/A Propellant Actuated Initiator, referred to herein as "initiator."

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

**2.2 Government documents.**

**2.2.1 Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-L-757 | Lead Styphnate |
| MIL-L-3055 | Lead Azide |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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| | |
|---|---|
| MIL-C-5604 | Compass; Magnetic, Pilot's Standby |
| MIL-D-21625 | Design and Evaluation of Cartridges for Cartridge Actuated Devices |
| MIL-P-20444 | Primer, Percussion, M42 Parts for and Loading, Assembling and Packing |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-453 | Inspection, Radiographic |
| MIL-STD-810 | Environmental Test Methods and Engineering Guidelines |
| MIL-STD-1168 | Ammunition Lot Numbering |

(Unless otherwise indicated, copies of Federal and Military specifications and standards are available from: Standardization Documents Order desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

**2.2.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

SPECIFICATIONS

Naval Sea Systems Command (CAGE Code 53711)

| | |
|---|---|
| WS 5003 | HNS Explosive |

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Data Review and Control Branch (Code 8410P), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

DRAWINGS

Naval Air Systems Command (CAGE Code 30003)

| | |
|---|---|
| DL 850AS131 | Initiator, Propellant Actuated, JAU-27/A |
| 850AS815 | Vibration Fixture, Initiator, Propellant Actuated, JAU-27/A |
| 850AS818 | Test Fixture: Initiator, Propellant Actuated, JAU-27/A |

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Data Review and Control Branch (Code 8410P), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

**2.3 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).



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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 545 Standard Method for Determining Image Quality in Direct Thermal Neutron Radiographic Examination

(Application for copies of ASTMs should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

**2.4 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.1.

**3.2 Primary components.** Only one lot of primary components shall be used in each lot of initiators, although, a primary component lot may be used in more than one initiator lot.

**3.2.1 OEA P/N 2912100-1.** The primary components for the OEA P/N 2912100-1 initiator are: the M42C1-PA101 percussion primer per MIL-P-20444, lead styphnate per MIL-L-757, lead azide per MIL-L-3055, and hexanitrostilbene (HNS) Type II, grade “A” per WS5003.

**3.2.1.1 M42C1-PA101 percussion primer.** The M42C1-PA101 Percussion Primer used in the OEA P/N 2912100-1 initiator shall be manufactured and tested in accordance with MIL-P-20444. The primers shall be loaded into the initiators within 24 months of new primer acceptance testing or within 12 months of primer recertification testing.

**3.2.2 SCOT P/N 6034100.** The primary components for the SCOT P/N 6034100 initiator are: the PVU-1/A percussion primer per MIL-P-20444, lead azide per MIL-L-3055, and the hexanitrostilbene (HNS) Type II, grade “A” per WS5003.

**3.2.2.1 PVU-1/A percussion primer.** The PVU-1/A Percussion Primer used in the SCOT P/N 6034100 initiator, shall be manufactured and tested in accordance with MIL-P-20444. The primers shall be loaded into the initiators within 24 months of new primer acceptance testing or within 12 months of primer recertification testing.

**3.3 Initiator production.** The initiators shall be manufactured in accordance with DL 850AS131 and the documents listed thereon. Each initiator shall meet the requirements of 3.3.1 through 3.3.1.3 and 3.3.2 through 3.3.2.5, respectively. Failure of any initiator to meet the applicable requirements of 3.3 shall result in rejection of that unit.



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**3.3.1 Explosive train assembly inspection.**

**3.3.1.1 Visual inspection.** All explosive train assemblies manufactured under contract shall be visually inspected and screened for defects as outlined in 4.3.3.

**3.3.1.2 Leakage.** Explosive train assemblies shall not exhibit a leak rate in excess of $1 \times 10^{-5}$ cubic centimeters per second ($cm^3/s$) of air when tested in accordance with 4.3.4.

**3.3.1.3 Radiographic inspection.** Each explosive train assembly shall be examined radiographically in accordance with 4.3.5.1. Presence of the propellant, non-metal parts, no foreign hydrogenous materials and the proper assembly of pyrotechnic/non-metallic materials will be ensured. Any observable imperfections in assembly will be cause for rejection.

**3.3.2 Initiator inspection.**

**3.3.2.1 Firing pin.** Free movement, proper alignment and protrusion of the firing pin shall be verified during assembly of each firing mechanism produced under contract in accordance with 4.3.1.

**3.3.2.2 Firing mechanism.** Firing mechanisms from each lot, selected in accordance with 4.3.2, shall meet the requirements listed below when subjected to the test prescribed in 4.3.2.1 and 4.3.2.2.

**3.3.2.2.1 Non-actuation.** Firing mechanisms shall not actuate when a pressure of 400 −25, +0 psig is applied for 1 minute at the inlet port. See 4.3.2.1.

**3.3.2.2.2 Actuation.** Firing mechanisms not previously subject to the non-actuation pressure in accordance with 3.3.2.2.1 shall actuate when a pressure of 600 +25, -0 psig is applied (see 4.3.2.2).

**3.3.2.3 Visual inspection.** All initiators manufactured under contract shall be visually inspected and screened for defects as outlined in 4.3.3.

**3.3.2.4 Radiographic inspection.** Initiators shall show proper assembly, presence of parts, and sealing when examined radiographically in accordance with 4.3.5.2.

**3.3.2.5 Demagnetization.** Initiators shall not deflect an approved Pilot's standby compass more than three degrees in either direction when passed in front of the compass. The initiators shall be tested in accordance with 4.3.6.

**3.4 Initiator performance.** Initiators shall be capable of meeting all requirements of 3.4.1 through 3.4.4.4.

**3.4.1 Vibration.** There shall be no evidence of deformation or derangement of components. The initiator shall not autoignite, shall be safe to handle, and shall meet the requirements of 3.3.2.4 and 3.4.4 following vibration testing of 4.3.7.

**3.4.2 Temperature and humidity (T & H) cycling.** The initiators shall meet the requirements of 3.3.2.4 and 3.4.4 after being exposed to the T & H cycle in accordance with 4.3.8.



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**3.4.3 Shock.** The initiators shall meet the requirements of 3.3.2.4 and 3.4.4 following shock testing of 4.3.9.

**3.4.4 Ballistic requirements.** Initiator(s) shall meet the requirements of 3.4.4.1 through 3.4.4.4 when tested in accordance with 4.3.10.

**3.4.4.1 Shear pin actuation pressure.** The upper (U) and lower (L) limits for shear pin actuation pressure (psig) when fired over the temperature range from –65°F to +200°F shall be U= 600 psig and L = 400 psig in accordance with 4.3.10.1.

**3.4.4.2 Actuation time.** The limit for actuation time (milliseconds) shall not be greater than 10 milliseconds when evaluated in accordance with 4.3.10.2.

**3.4.4.3 High order detonation.** The minimum plate dent for high order detonation of the output SMDC test tip on the aluminum witness plate shall be 0.001 inch when evaluated in accordance with 4.3.10.3.

**3.4.4.4 Misfire.** There shall be no misfires (see 4.3.10.4).

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Production inspection (see 4.2.2)
c. Lot acceptance inspection (see 4.2.3)

**4.2 Types of inspections.**

**4.2.1 First article inspection**. Before entering into quantity production, an acceptable first article sample of 54 initiators shall be prepared using the same procedures and methods proposed for the production lot. A first article sample acceptable for functional and environmental testing shall be defined as a sample which has met the applicable requirements of 3.2 through 3.4 of this specification. All fifty-four initiators shall be expended in the tests (see table I below). The contractor shall deliver the initiators to the activity designated by the contracting agency for tests. Any damage inflicted by the environmental treatments, which would adversely affect the performance of the item in service application shall cause rejection of the first article sample. Any further production prior to notification by the contracting agency of first article sample acceptability shall be at the contractor's risk. Failure of any initiator to comply with the requirements of section 3 shall be cause for rejection of the first article represented.

**4.2.2 Production inspection.** All production initiators manufactured under contract shall be inspected and screened to requirements specified in 3.2 through 3.3. Any unit failing to meet the requirements listed in 3.2 through 3.3 shall be rejected and removed from the lot.

**4.2.3 Lot acceptance inspection.** Lot acceptance inspection shall consist of the examinations and tests specified in table II. Failure of any sample initiator to comply with the requirements listed in table II shall be cause for rejection of the lot represented.



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TABLE I. First article test plan.

| Test Sequence | Requirement para. | Test paragraph | Sample Group | | | |
|---|---|---|---|---|---|---|
| | | | I | II | III | IV |
| Visual | 3.3.2.3 | 4.3.3 | 6 | 18 | 6 | 24 |
| Radiographic | 3.3.2.4 | 4.3.5 | 6 | 18 | 6 | 24 |
| Vibration | 3.4.1 | 4.3.7 | 6 | | | |
| T & H Cycling | 3.4.2 | 4.3.8 | | 18 | | |
| Shock | 3.4.3 | 4.3.9 | | | 6 | |
| Radiographic | 3.3.2.4 | 4.3.5 | 6 | 18 | 6 | |
| Ballistic test<br>-65 ± 5°F<br>70 ± 5°F<br>200 ± 5°F | 3.4.4 | 4.3.10 | <br>2<br>2<br>2 | <br>6<br>6<br>6 | <br>2<br>2<br>2 | <br>8<br>8<br>8 |

TABLE II. Lot acceptance inspections and tests.

| Inspection/Test | Test Paragraph | Quantity |
|---|---|---|
| Visual | 4.3.3 | All |
| Radiographic | 4.3.5 | All |
| Ballistic test<br>-65 ± 5°F<br>70 ± 5°F<br>200 ± 5°F | 4.3.10 | <br>1/3 test sample<br>1/3 test sample<br>1/3 test sample |

**4.2.3.1 Sample size.** A random sample from each production lot, including the retained sample for investigative purposes, shall be selected in accordance with table III for lot acceptance inspection. Test sample initiators and initiators retained for investigative purposes shall not be applied as part of the quantity specified for delivery by the contract or purchase order.

TABLE III. Lot acceptance sampling.

| Lot Size | Test Sample Size | Retained Sample Size |
|---|---|---|
| 2 to 50 | 9 | 3 |
| 51 to 100 | 12 | 3 |
| 101 to 150 | 21 | 3 |
| 151 to 280 | 33 | 3 |
| 281 to 500 | 51 | 3 |



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**4.3 Inspections and tests.**

**4.3.1 Firing pin assembly inspection.** The firing pin used in each initiator manufactured under contract shall be subjected the tests described in 4.3.1.1 and 4.3.1.2 prior to and during assembly of the firing mechanism. Parts not satisfying these requirements shall be discarded.

**4.3.1.1 Firing pin movement.** The firing pin, without O-ring, shall move freely under its own weight throughout maximum travel provided for the pin in the firing mechanism.

**4.3.1.2 Alignment.** Alignment and protrusion of the firing pin striker shall be verified.

**4.3.2 Firing mechanism test.** From each lot, a number of firing mechanisms shall be selected in accordance with Table I, General Inspection Level I, of ASQC Z1.4 with the additional requirement that the sample size used for non-actuation and actuation testing shall not be less than eight units. If the sample is not equally divisible, the excess initiator shall be placed in the group to be tested at -65°F. The firing mechanism will be mated to an inert explosive train cartridge to perform testing.

**4.3.2.1 Non-actuation.** Half of the samples chosen for this test shall be conditioned for at least 4 hours but not more than 24 hours at -65°F ± 5°F. The remaining half shall be conditioned for at least 4 hours but not more than 24 hours at 200°F ± 5°F. Within three minutes after removal from the conditioning chamber a pressure of 400 +0, −25 psig shall be applied to the inlet port of each firing mechanism for 1 minute. Actuation of any firing mechanism shall result in rejection of the lot.

**4.3.2.2 Actuation.** Samples will be chosen as outlined in paragraph 4.3.2. Half the samples shall be conditioned to -65°F ± 5°F for at least 4 hours but not more than 24 hours. The other half of the samples shall be conditioned to +200°F ±5°F for at least 4 hours but not more than 24 hours. Within three minutes after removal from the conditioning chamber, a pressure of 600 +25, -0 psig shall be applied to the inlet port of each firing mechanism. Failure of any firing mechanism to actuate shall result in rejection of the lot. Proper actuation is defined as the shearing of the shear pin, followed by free travel and bottoming of the firing pin against the head of an inert cartridge.

**4.3.2.3. Rework.** If the cause of failure in 4.3.2.1 and 4.3.2.2 can be corrected by rework of the lot, the lot in question may be reworked to correct the deficiency and a new sample, selected in accordance with Table I, General Inspection Level I, of ASQC Z1.4, submitted for test. Firing mechanisms which pass the tests of 4.3.2.1 and 4.3.2.2 may be reworked by replacement of the shear pins, O-rings, and damaged firing pins.

**4.3.3 Initiator and explosive train assembly visual inspection.** Visually examine all explosive train assemblies prior to initiator assembly. Each assembled initiator shall be visually examined. Reject those with inadequate or illegible descriptive markings or serial numbers, dents, deep scratches, or other defects which may prevent proper installation or function.

**4.3.4 Explosive train assembly leak test.** Each explosive train assembly produced under contract shall be leak tested in a dry gas leak tester to determine conformance with 3.3.1.2. Units which exhibit a leak rate in excess of $1 \times 10^{-5}$ standard $cm^3$/sec air at a pressure differential of 1.0 ± 0.1 atmosphere shall be rejected.



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**4.3.5 Radiographic examination.**

**4.3.5.1 Explosive train assemblies examination.** Radiographic examination of explosive train assemblies shall include X-ray in accordance with MIL-STD-453 and N-ray in accordance with ASTM E 545. All explosive train assemblies shall be identified with serial numbers beginning with 001 prior to examination. The initiators shall be arranged on trays or boards in consecutive numerical order with any missing serial numbers identified on the radiographic plate. The explosive train assemblies shall be positioned for the most revealing exposure with the long axis perpendicular to the radiographic beam. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4 × 6 inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, the DODIC, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the tests (see 6.2). Defective assemblies found during radiographic review are to be marked on the radiographic plate and removed from the production lot.

**4.3.5.2 Initiator examination.** Radiographic examination of initiators shall include x-ray in accordance with MIL-STD-453. All initiators shall be identified with serial numbers beginning with 001 prior to examination. The initiators shall be arranged on trays or boards in consecutive numerical order with any missing serial numbers identified on the radiographic plate. The initiators shall be positioned for the most revealing exposure with the long axis perpendicular to the radiographic beam. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4 × 6 inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, the DODIC, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the tests (see 6.2). Defective initiator assemblies found during radiographic review are to be marked on the radiographic plate and removed from the production lot.

**4.3.6 Demagnetization.** Each initiator shall be moved past a pilot's standby compass face (in accordance with MIL-C-5604) by placing the initiator 6 inches from the compass set in an east-west heading. The initiator and the compass shall be placed at the same height from the floor. The initiators shall be moved past the compass face in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about it's longitudinal axis, and moved back past the compass face. This procedure shall be repeated for each 90 degrees rotation of the initiator. The initiator shall be acceptable if the requirements of 3.3.2.5 are met.

**4.3.7 Vibration test.**

**4.3.7.1 Non-gunfire vibration.** The test shall be conducted in accordance with MIL-STD-810B, Table 514.1-I Curve "AT" except that for each cycling period the time shall be divided equally for vibration at −65, 70, and 200°F. The initiators shall be mounted in the vibration test fixture, Drawing 850AS815, for the JAU-27/A initiator. The non-gunfire vibration test shall be performed on all six initiators. The six initiators shall be mounted on the test fixture at the same time and one shall be monitored for response during the test.

**4.3.7.2 Gunfire vibration.** The gunfire vibration test shall be performed on all six initiators, on each of the three orthogonal axes, and shall be conducted at 70°F test temperature only. The six initiators shall be mounted on the test fixture at the same time and one shall be monitored for response during the test.



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**4.3.7.2.1 Resonance search.** A resonance survey shall be conducted on each of the six initiators at 70°F in each of the three orthogonal axes. The frequency sweep shall be made from 50 Hz to 500 Hz and back to 50 Hz in approximately 7.50 minutes to the vibration test levels shown in table IV. The resonance points shall be noted. A resonance point is defined as a frequency which has a transmissibility (output/input) of two or greater. If more than three resonant points are noted from the resonance survey, only the three more severe will be used for resonance dwell vibration.

TABLE IV. Vibration test levels.

| Frequency (Hz) | Acceleration or Amplitude |
|---|---|
| 50 to 54 | 0.36 inch D.A. |
| 55 to 210 | ± 5 G's |
| 211 to 300 | 0.00222 inch D.A. |
| 301 to 500 | ± 10 G's |

**4.3.7.2.2 Resonance dwell.** The six initiators shall be vibrated in each of three orthogonal axes at 70°F for 7.50 minutes at each resonant point (to a maximum of three) obtained during the resonance survey in 4.3.7.2.1 herein. The gunfire vibration test levels are shown in table IV depending upon the number of resonance frequencies determined in the resonance survey per 4.3.7.2.1. The remainder of the sweep time in accordance with table V shall be used for vibration cycling per 4.3.7.2.3 herein.

**4.3.7.2.3 Vibration cycling.** All six initiators shall be subjected to sinusoidal vibration cycling at 70°F for a period of time as specified in 4.3.7.2.2 on each of the three orthogonal axes with the frequency varying logarithmically from 50 Hz to 500 Hz and back to 50 Hz in approximately 7.50 minutes. The gunfire vibration test levels shall be shown in table IV.

**4.3.7.2.4 Fixed dwell.** After sinusoidal vibration cycling, the six initiators shall be subjected to a series of fixed dwells as shown in table V. The fixed narrow band dwells shall be performed by sweeping the frequency of ± 5 percent of the specified center frequency (e.g., the 100 Hz dwell should be performed from 95 to 105 Hz at the logarithmic sweep rate specified in 4.3.7.2.3).

**4.3.7.2.5 Sinusoidal sweep.** After completion of the fixed dwell testing, the six initiators shall be subjected to seven sinusoidal vibration sweeps in each of the three orthogonal axes. The frequency shall vary logarithmically from 500 Hz to 2000 Hz and back to 500 Hz. The sweep cycling rate shall be as specified in 4.3.7.2.3 herein, with an applied input of ± 10 G's.

**4.3.8 Temperature and humidity cycling.** The temperature and humidity cycling test shall be conducted in accordance with MIL-D-21625F. Failure of any cartridge to comply with the requirements of 3.4.2 shall be cause for rejection of the lot or first article.

**4.3.9 Shock test.** The shock test shall be conducted in accordance with MIL-D-21625F. Failure of any initiator to comply with the requirements of 3.4.3 shall be cause for rejection of the lot or first article.



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TABLE V. Gunfire sinusoidal vibration test schedule

| Fixed Dwell (fd) at the following center frequencies | Number of Resonances | | | |
|---|---|---|---|---|
| | 0 | 1 | 2 | 3 |
| | Sweep Time | | | |
| | 30 | 22.5 | 15.0 | 7.50 |
| | Resonance Dwell Time | | | |
| | – | 7.5 | 15.0 | 22.50 |
| 67 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 100 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 135 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 200 Hz | 15.00 | 15.00 | 15.00 | 15.00 |
| 267 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 300 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 335 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 400 Hz | 15.00 | 15.00 | 15.00 | 15.00 |
| 467 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 500 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| Total Time Each Axis | 120.0 | 120.0 | 120.0 | 120.0 |

Note: When an equipment resonance occurs within 5% of a fixed dwell frequency, the fixed dwell period shall be omitted and only the resonance dwell performed. The omitted fixed dwell time shall then be added to the sweep time.

**4.3.10 Ballistic test.** Sample initiators shall be conditioned for at least 4 hours but not more than 24 hours at the specified temperatures. The initiators shall be removed from the conditioning chamber and test fired within 3 minutes. If any initiator is not fired within 3 minutes after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested. The initiators shall be fired in the test fixture as shown in figure 1(see Drawing 850AS815). Instrumentation shall be provided to measure and record the input air pressure (800 +25,-0 psig); shear pin actuation pressure (psig); actuation time (milliseconds); and the high order detonation of the output shielded mild detonating cord (SMDC) test tip on the aluminum witness plate. The initiators shall meet the requirements specified in 3.4.4.

**4.3.10.1 Shear pin actuation pressure.** The shear pin actuation pressure is the gas pressure required to actuate the gas operated firing mechanism when a gas pressure of 800 +25, −0 psig is applied at a rate between 10,000 psig/sec and 50,000 psig/sec. The lot shall be acceptable with respect to shear pin actuation pressure if the ballistic sample meets the requirements of 3.4.4.1.



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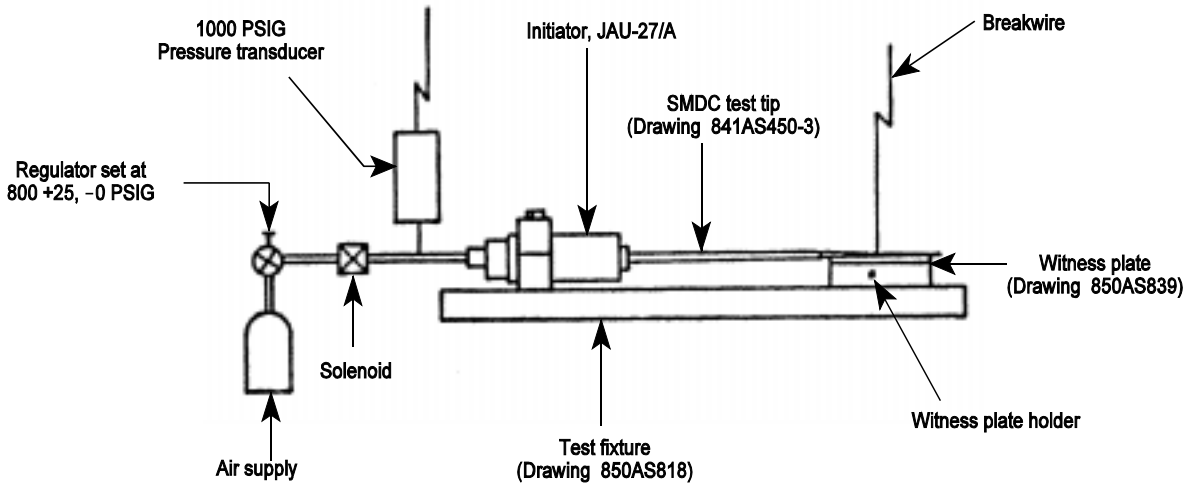



FIGURE 1. Test setup.

**4.3.10.2 Actuation time.** The actuation time shall be measured from the initiation of the inlet air pressure supply to the detonation of the outlet SMDC test tip. Each initiator shall meet the requirements of 3.4.4.2.

**4.3.10.3 High order detonation.** High order detonation of the output SMDC test tip shall be demonstrated in a 6061-T6 Aluminum witness plate. Each initiator shall meet the requirements of 3.4.4.3.

**4.3.10.4 Misfire.** An initiator that does not fire under the conditions specified in 4.3.10 shall be considered defective, and the lot shall be considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure. Each initiator shall meet the requirements of 3.4.4.4.

**4.3.10.5 Test failure.** If test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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## 6. NOTES

**6.1 Intended use.** The initiator, propellant actuated converts ballistic gas energy into the explosive energy required to initiate the shielded mild detonating cord (SMDC) used in the FA-18 A/B/C/D/E/F, F-14 and TAV-8B aircraft.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).

c. Whether first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2.1).

d. Directions for shipping radiographs of entire production lot and marriage log (report indicating cartridge serial number to initiator serial number), along with the ballistic sample to the activity conducting the production lot acceptance tests (see 4.3.3.1 and 4.3.3.2).

e. That the safety requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable.

**6.3 First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

**6.4 Subject term (keyword) list.**

Initiator
Percussion
Powder
Primer
Propellant

Preparing Activity
Navy - OS
(Project 1377-0053)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER**<br>MIL-DTL-82960(OS) | **2. DOCUMENT DATE (YYMMDD)**<br>961023 |
|---|---|---|

INITIATOR, PROPELLANT ACTUATED, JAU-27/A (MF72)

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include Zip Code)* | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>*(if applicable)* | **7. DATE SUBMITTED**<br>(YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE *Include Area Code)* | |
|---|---|---|
| Commander, Indian Head Division<br>Naval Surface Warfare Center | (1) Commercial<br>(301) 744-4700 | (2) AUTOVON<br>354-4700 |
| Standardization Team (Code 840M)<br>101 Strauss Avenue<br>Indian Head, MD 20640-5035 | Defense Quality and Standardization Office (DLSC-LM)<br>8725 John J. Kingman Road, Suite 2533<br>Fort Belvoir, Virginia 22060-6221<br>(703) 767-6888 DSN: 427-6888 | |



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